

FR-016 M2 — Deterministic Supervisor Runtime and Go/No-Go Evaluation

FR016 — sources/optional/m2.md

Masterclass source package snapshot

Functional Requirement	FR-016
Edition	Engineering Learning Academy — Source Package snapshot
Status	Complete (M2) — **GO AS LEARNING PROOF ONLY** — **historical milestone record
Audience	Academy generation and study (snapshot; SoT remains authoritative)
Document type	Source package snapshot (PDF)

Faithful rendering of a regenerable Masterclass source snapshot. Do not hand-edit; regenerate via python scripts/build_masterclass_package.py. Repository documentation remains the source of truth.

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FR-016 M2 — Deterministic Supervisor Runtime and Go/No-Go Evaluation

Date: 2026-08-06

Status: Complete (M2) — **GO AS LEARNING PROOF ONLY** — historical milestone record

Succeeded by: [M3](#); [M4](#); [acceptance](#)

Architecture: [ADR-008](#)

Preceding: [M1 contracts](#)

Did not begin in M2: M3 CLI productisation, FR-017, job discovery, frameworks

1. Executive summary

M2 delivered the smallest deterministic multi-agent runtime for the accepted topology:

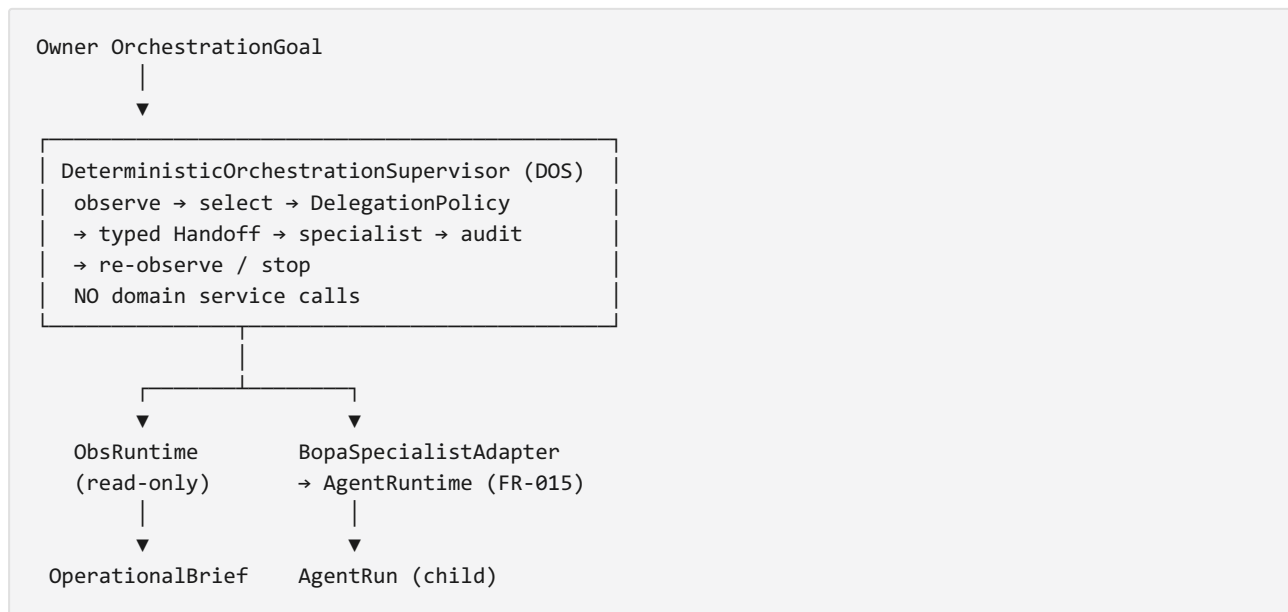
DOS (delegates only) → **BOPA** (frozen mutating specialist via thin adapter) and/or **OBS** (strictly read-only operational briefing).

Corpus **A-O: 15/15 PASS**. Manual runner (`scripts/run_fr016_m2_manual.py`) reconstructs specialist selection, handoffs, authority, results, stop reasons, and owner actions.

Go/no-go: GO AS LEARNING PROOF ONLY

- Permission separation, typed handoffs, loop controls, and audit work.
 - OBS removes a real owner interpretation task on pipeline-advises and truth-blocked cases without broadening BOPA.
 - Near-term commercial value remains **modest**: for happy-path prepare, direct `cic agent` is still simpler.
 - Proceed to a **minimal M3/M4** only to close the learning milestone and freeze documentation — **do not claim product value**. Defer richer productisation until Job Discovery (or another genuine specialist boundary).
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2. Runtime architecture



Package: `career_intelligence.multi_agent`

Persistence: in-memory (tests) / `data/orchestration_runs/` (JSON store)

BOPA package: **unchanged**.

3. DOS behaviour

Step	Behaviour
Observe	<code>ObservationBuilder</code> → <code>OrchestrationObservation</code> (+ hash)
Select	<code>select_next_specialist</code> (deterministic)
Delegate	<code>evaluate_delegation_policy</code> sole admission
Handoff	Append-only lifecycle pending → accepted → executing → completed/stopped
Specialist	OBS or BOPA only
Stop	Explicit <code>OrchestrationStopReason</code> ; owner action text when awaiting
Forbidden	<code>attempt_domain_work()</code> raises <code>DomainWorkForbiddenError</code>

Default limits: max steps 12; max visits/specialist 3; no-progress / repeated / circular delegation enforced.

4. BOPA adapter

`BopaSpecialistAdapter` maps:

`Handoff` → `AgentGoal(prepare_for_owner_review)` → `AgentRuntime.start|resume` →
`BopaSpecialistResult` (child `AgentRun` id + stop reason).

Does **not** change allow-list, ToolPolicy, material-benefit, truth, or owner gates. DOS cannot impersonate BOPA tools.

5. OBS behaviour and example briefs

OBS allow-list remains read-only. `ObsRuntime` policy-gates inspect actions then `compose_brief` → `OperationalBrief`.

Example — interviewing (pipeline advises):

- needs: `pipeline_advises_against_preparation`
- next: `owner_review`
- summary: review pipeline before any preparation
- pipeline note: preparation usually unnecessary while interviewing
- **BOPA not selected**

Example — truth blocked:

- needs: `truth_blockers_need_synthesis` (+ brief-only if applicable)
 - next: `owner_remediate_truth`
 - blockers listed; no waive/rewrite
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6. Delegation policy

Unchanged core from M1, used live by DOS:

- brief goals → OBS only
- coordinate + pipeline/ambiguity/truth → OBS first
- coordinate + missing package (no brief-first signals) → BOPA
- `synthesize_after_prepare` → OBS after BOPA once
- visit / step / circular / repeated / approval gates

Owner notes never affect policy inputs (injection-safe).

7. Handoff lifecycle

Persisted fields include `ids`, `source=supervisor`, `target`, `opportunity`, `goal`, `state hash`, `policy result`, `child AgentRun/brief refs`, `timestamps`, `acceptance`.

Acceptance values: `pending` → `policy_blocked` | `accepted` → `executing` → `completed` | `stopped` (also `rejected` / `stale` / `cancelled`).

8. Orchestration audit / store

`OrchestrationRun` holds `goal`, `status`, `specialist visits`, `handoff ids`, `child BOPA ids`, `brief id`, `budgets`, `stop reason`, `owner action`, `append-only events`.

Stores: `InMemoryOrchestrationStore`, `JsonDirectoryOrchestrationStore` (`data/orchestration_runs/`, `handoffs/`, `briefs/`).

Does **not** replace `AgentRun` or domain `SoTs`.

9. Loop / deadlock controls

Control	Evidence
Max global steps	Deny → <code>orchestration_max_steps</code>
Max visits / specialist	Deny → <code>specialist_visit_limit</code>
Repeated handoff	Corpus I
Circular sequence	Corpus J
No-progress	Unchanged observation hash rounds
Fail-closed	Provider unavailable → stop (Corpus M)

10. Checkpoint / resume

- Full run persisted each step (`checkpoint_ref`).
- `resume` requires `awaiting_owner` or `running`.
- Re-observes authoritative/static state; rejects stale hashes via new observe.
- OBS skips regenerating brief when hash unchanged.
- BOPA adapter resumes incomplete child `AgentRun` when present.

- Corpus K/L pass.

11. Corpus results

ID	Case	Result
A	Brief-only	PASS
B	Preparation → BOPA → owner stop	PASS
C	BOPA → OBS synthesis	PASS
D	Pipeline advises → OBS not BOPA	PASS
E	Truth-blocked OBS brief	PASS
F	Illegal delegation	PASS
G	OBS mutate forbidden	PASS
H	DOS domain work forbidden	PASS
I	Repeated handoff	PASS
J	Circular sequence	PASS
K	Partial resume / no duplicate prep	PASS
L	Stale state re-inspect	PASS
M	Provider unavailable	PASS
N	Prompt injection	PASS
O	Pipeline safety	PASS

15/15 PASS (`career_intelligence.multi_agent.evaluation.run_corpus`).

12. Manual validation

`python scripts/run_fr016_m2_manual.py` — PASS (journeys 1–7 + corpus).

Owner can see: why specialist selected, authority, handoff, result, stop reason, next owner action.

13. Tests

Suite	Result
tests/unit/multi_agent/ (contracts + runtime)	green
tests/unit/agent/test_models.py + test_runtime.py	green (BOPA regression)
Corpus A–O	15/15
Manual runner	PASS

14. Value comparison vs direct owner commands

Scenario	Direct owner	DOS+OBS+BOPA	Verdict
Happy-path prepare	cic agent run	DOS→BOPA	Direct is simpler
Interviewing + apply + no package	Owner must notice pipeline on show	DOS→OBS brief-first	OBS helps
Truth blockers	cic agent show / truth CLI	OBS synthesises blockers	Slight help
Illegal submit intent in notes	ToolPolicy on BOPA	Delegation ignores notes	Safety parity
Audit across prep+brief	Separate AgentRun + CLIs	Parent OrchestrationRun	Clearer for learning

Owner commands reduced: small, mainly for brief-first routing.

Non-obvious DOS choices: yes (D, E, N).

OBS removes a meaningful task: yes for pipeline-advises / truth synthesis without mutate authority.

15. Complexity assessment

Layer	Cost
Contracts + policies	Justified (M1)
DOS runtime + stores	Moderate
OBS	Small and clear
BOPA adapter	Thin
Parent/child audit	Useful for FR-017 later
Product CLI (M3)	Not yet justified as product

Complexity is **proportionate as a learning proof, not** as a daily replacement for `cic agent` on happy paths.

16. Defects and technical debt

Item	Notes
StaticObservationBuilder queues	Test/corpus convenience; live builder uses readiness
Resume after completed brief	Stops with briefing_complete; no re-prep unless state changes
No wall-clock timeout yet	Step/visit budgets cover offline; wall clock deferred
Presentation is script-only	M3 CLI intentionally not started
<code>synthesize_after_prepare</code> flag	Explicit opt-in to avoid surprising OBS after every BOPA

No frozen BOPA behaviour changes. No FR-008–FR-015 reopen.

17. Mandatory go/no-go recommendation

Answers to M2 gate questions

Question	Answer
Does DOS add value beyond <code>cic agent</code> / reporting?	Yes, narrowly — non-obvious OBS vs BOPA routing and parent audit. Happy-path prepare: no .
Does OBS remove a meaningful owner task?	Yes — pipeline-advises and truth-blocker synthesis without mutate tools.
Do handoffs + separated permissions improve safety/audit/extensibility?	Yes — corpus F/G/H/I/J/N/O; clean child AgentRun linkage.
Is complexity proportionate?	For learning/substrate: yes. For near-term product: no.
Continue / learning-proof / defer / reject?	GO AS LEARNING PROOF ONLY

Binding decision

2. GO AS LEARNING PROOF ONLY — complete minimal M3/M4 to close the milestone and freeze docs; **do not claim product value**. Do not expand specialists. Revisit product ambition when Job Discovery (or another real permission boundary) exists.

Not chosen: full GO (overclaims commercial value); NO-GO/DEFER (would abandon clean learning close-out after a working proof); REJECT (runtime is justified as proof).

18. Final repository status

Item	Status
M0 spike	Accepted with revisions
M1 contracts + ADR-008	Complete
M2 DOS + OBS + BOPA adapter	Complete
Corpus / manual / tests	Green
Go/no-go	GO AS LEARNING PROOF ONLY
M3	Not started (owner request required; minimal only)
FR-017	Not started
BOPA frozen behaviour	Unchanged

Owner next step

Acknowledge the **GO AS LEARNING PROOF ONLY** decision, then optionally request minimal M3 (thin owner surface + docs) / M4 freeze — without product marketing claims.